1. Preprocessing: Normalization

Before the algorithm starts, you must normalize the data.

- **Zero Mean:** You subtract the mean from each feature so the data is centered at the origin (0,0).
- **Feature Scaling:** If x_1 is a huge number (like house size in square feet) and x_2 is a small number (like number of bedrooms), one feature will dominate the other. You scale them so they have a similar range.

2. Finding the New Axis (The Principal Component)

The goal is to find a new axis (let's call it the **z-axis**) that best captures the information in the data.

- **Projection:** PCA tests different directions for this new axis. For every data point, it draws a line perpendicular (at 90 degrees) to the new axis. This is called **projecting** the data.
- **Maximizing Variance:** How does it choose the best axis? It looks for the direction that keeps the data points **spread out** as much as possible.
  - **Bad Choice:** If the projected points are squished together, you have lost information (low variance).
  - **Good Choice:** If the projected points are widely spread out, you have retained the maximum amount of information (high variance).

This best axis is called the **Principal Component**.

3. The Math: Dot Products

Once PCA finds the direction of the z-axis, it defines it using a "unit vector" (a vector of length 1). Let's say this vector is $[0.71, 0.71]$.

To convert a data point (e.g., $x_1 = 2, x_2 = 3$) into the new feature z:
$z = \text{Vector}_{data} \cdot \text{Vector}_{axis}$
$z = [2, 3] \cdot [0.71, 0.71] = 3.55$

Your new feature z is **3.55**. You have successfully reduced 2 dimensions into 1.

4. What if you want more dimensions?

If you have 50 features and want to reduce them to 2 (instead of 1), PCA finds the second axis (z_2) by looking for a direction that is:

1. **Perpendicular (90 degrees)** to the first axis (z_1).
2. Captures the **next highest** amount of variance.

5. PCA is NOT Linear Regression

It is easy to confuse the two, but they do completely different things:

Feature	Linear Regression	PCA
Type	Supervised Learning (uses labels y)	Unsupervised Learning (no labels)
Goal	Predict y given x	Reduce features / Visualization
Error Calculation	Minimizes **Vertical** distance (squared error between point and line)	Minimizes **Perpendicular** distance (distance from point to the axis)

6. Reconstruction

You can also go backward. If you have the value $z = 3.55$, you can try to recreate the original x_1 and x_2.

- **Formula:** $z \times \text{Vector}_{axis}$
- **Result:** $3.55 \times [0.71, 0.71] = [2.52, 2.52]$